

Clock generator & Reset fanout

File: clock_gen.kicad_sch

Direct 4-lane PCIe

File: direct_PCIe.kicad_sch

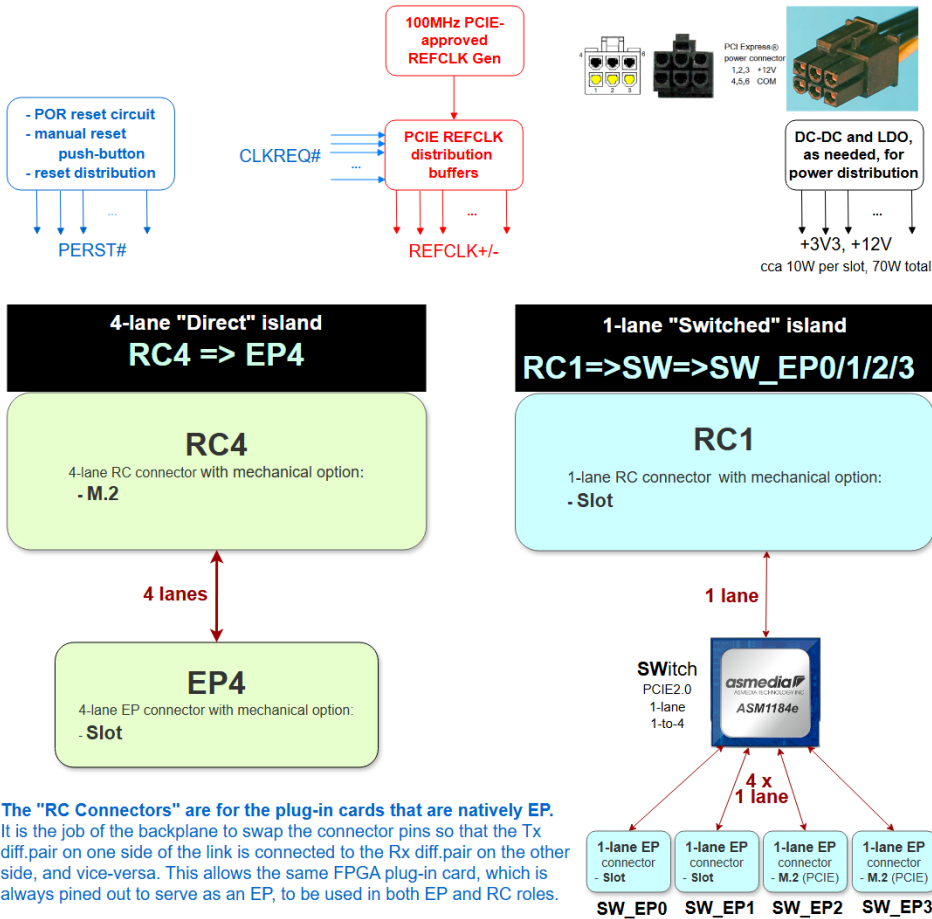
PCIe switch

File: pcie_switch.kicad_sch

Switched 4 x 1-lanes PCIe

File: 4_1-lanes_PCIe.kicad_sch

PCIe mini Backplane

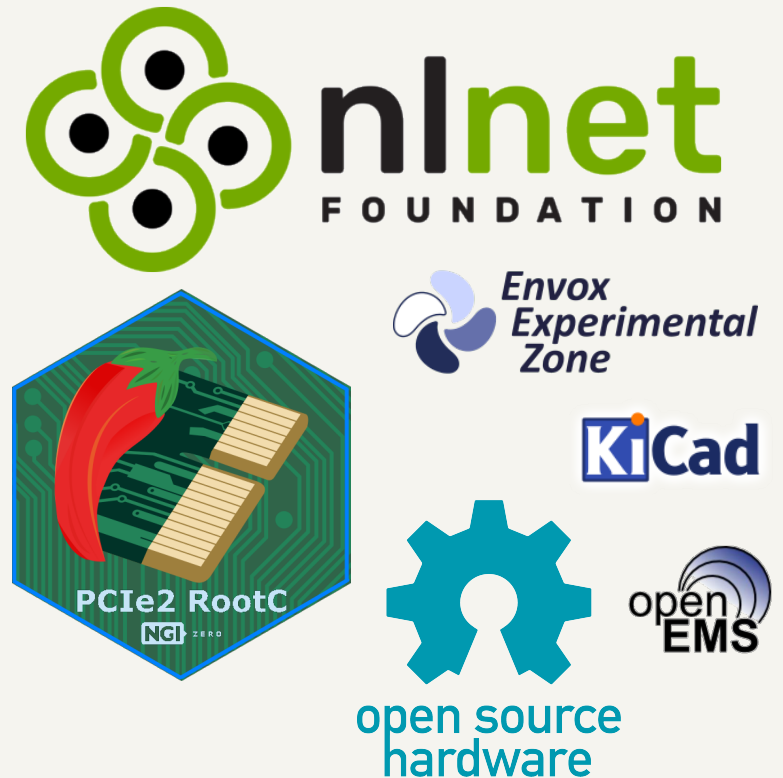


APPROVALS MAY BE VIEWED IN PLM SYSTEM

REV	ECO	DESCRIPTION	DRAWN BY	APPROVED BY	DATE

ELECTRONIC APPROVALS:

ALL APPROVALS FOR THIS DRAWING, ASSOCIATED PARTS LIST(S), AND REVISION CHANGES, ARE CONTROLLED IN THE PRODUCT



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Sheet: /
File: openpci2-backplane.kicad_sch

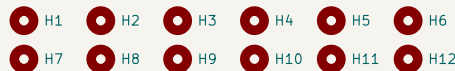
Title: openPCIe2 Backplane

Size: A4 Date: 2026-09-08

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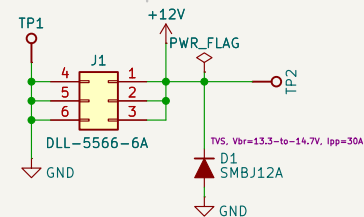
Rev: r1B1

Id: 1/5

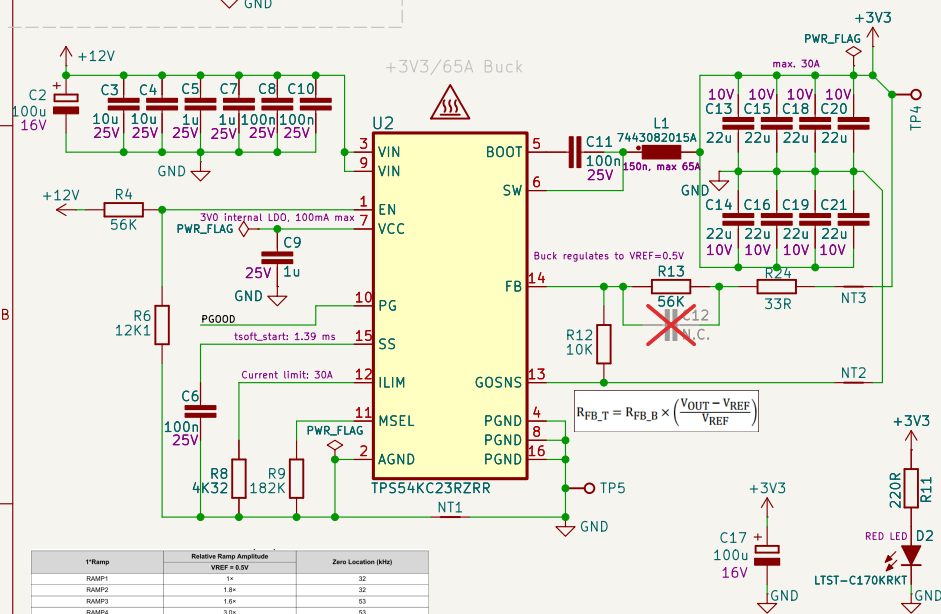


Power Supply

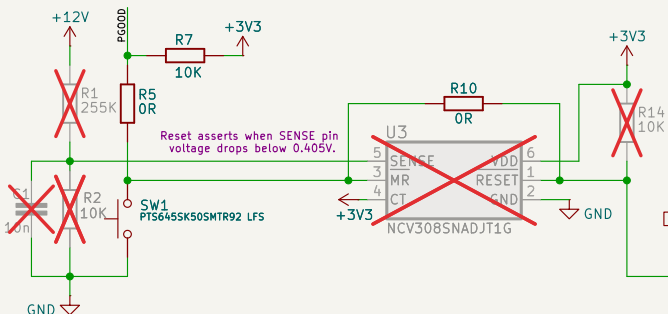
+12V Power input



+3V3/65A Buck

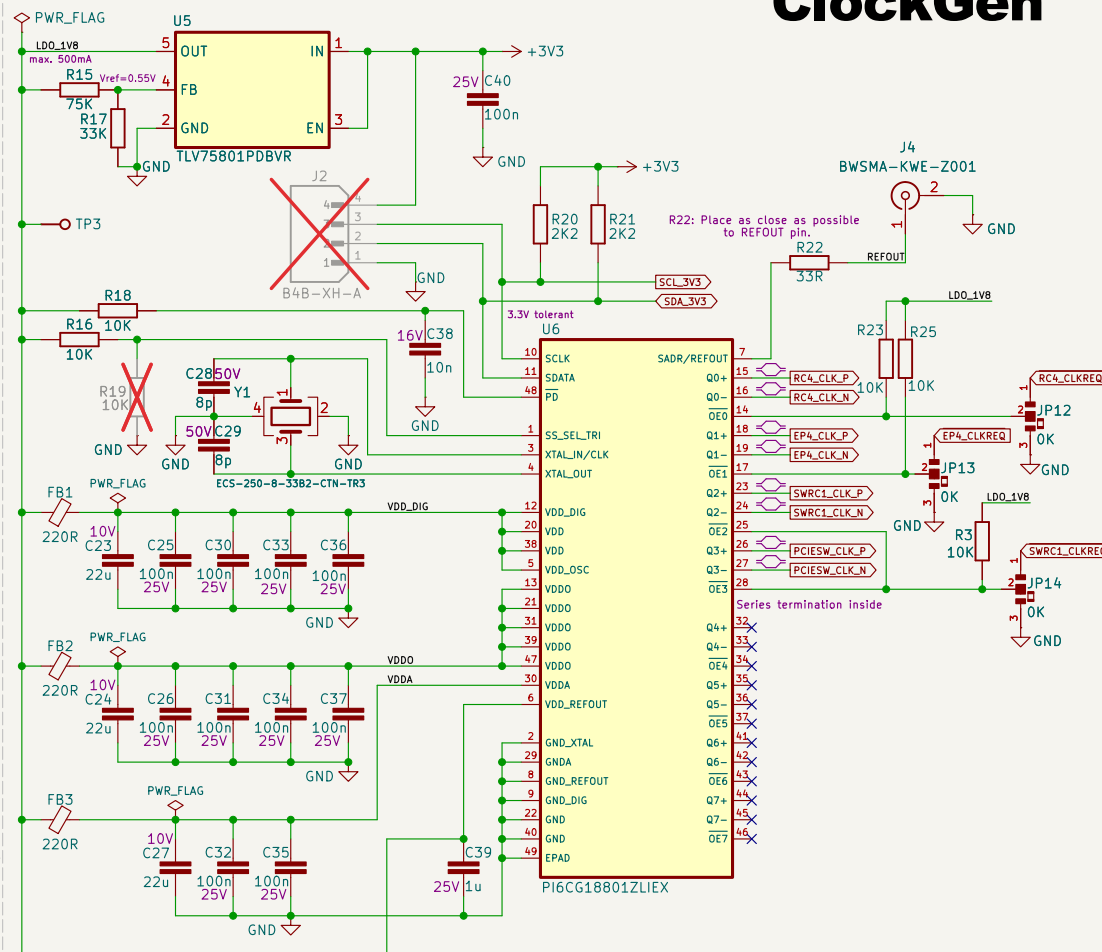


Reset asserts when +12V input drops below -10.73V.



Voltage Supervisor Manual Reset, Reset Fanout

ClockGen



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Sheet: /Clock generator & Reset fanout/
 File: clock_gen.kicad_sch

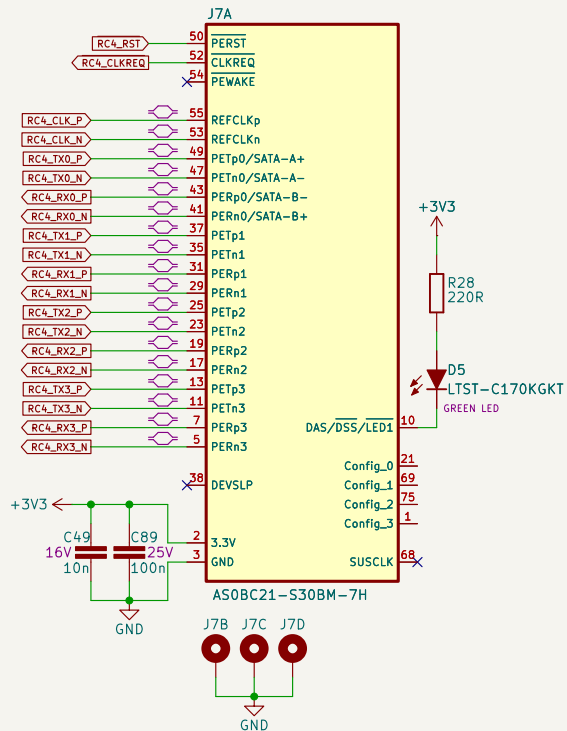
Title: openPCIE2 Backplane

Size: A4
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Date: 2026-09-08

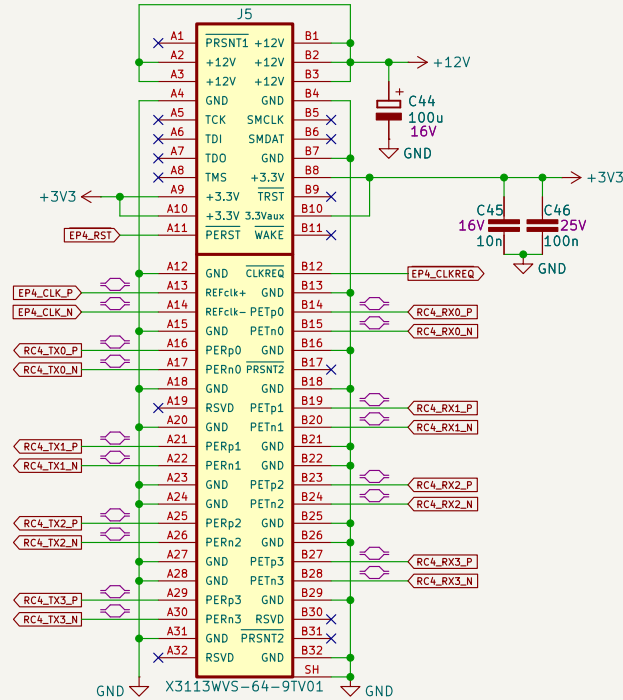
Rev: r1B1
 Id: 2/5

RC4



R and Connector must be placed right next to the trace to eliminate the stub.

EP4



R and Connector must be placed right next to the trace to eliminate the stub.

PCIEx1 pinout

Pin	Side B	Side A	Description
1	+12 V		Must connect to farthest PRSNT2# pin
2	+12 V	+12 V	Main power pins
3	+12 V	+12 V	
4	Ground	Ground	
5	Ground	Ground	SMBus and JTAG port pins
6	Ground	Ground	
7	Ground	Ground	
8	+3.3 V	+3.3 V	Aux power & Standby power
9	+3.3 V aux	+3.3 V	
10	+3.3 V aux	+3.3 V	
11		PERST#	Link reactivation, fundamental reset [25]
Key notch			
12	CLKREQ#(20)	Ground	Clock Request Signal
13	Ground	REFCLK+	Reference clock differential pair
14	HSOp(0)	REFCLK-	Lane 0 transmit data, + and -
15	HSOn(0)	Ground	
16	Ground	HSIp(0)	Lane 0 receive data, + and -
17	Ground	HSIn(0)	
18	Ground	Ground	PCI Express x1 cards end at pin 18

Legend

Ground pin	Zero volt reference
Power pin	Supplies power to the PCIe card
Card-to-host pin	Signal from the card to the motherboard
Host-to-card pin	Signal from the motherboard to the card
Open drain	May be pulled low or sensed by multiple cards
Sense pin	Tied together on card
Reserved	Not presently used, do not connect

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Sheet: /Direct 4-lane PCIe/

File: direct_PClie.kicad_sch

Title: openPCIe2 Backplane

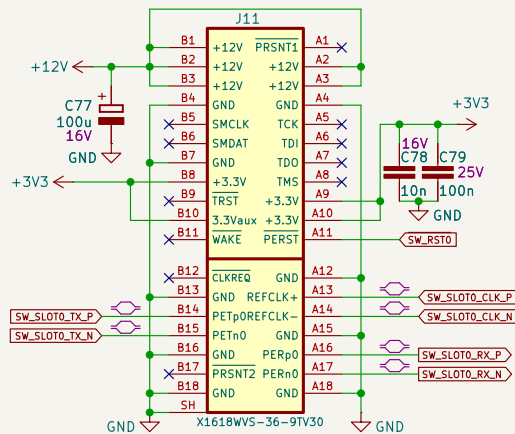
Size: A4 Date: 2026-09-08

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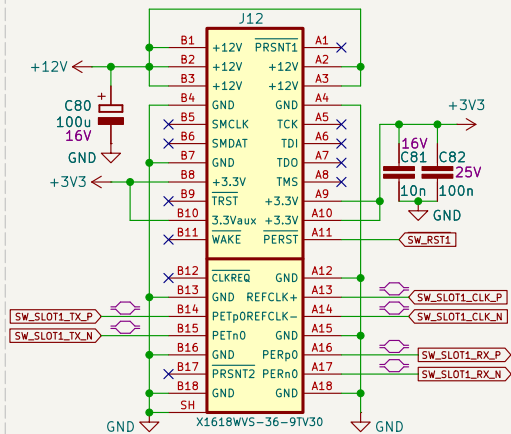
Rev: r1B1

Id: 3/5

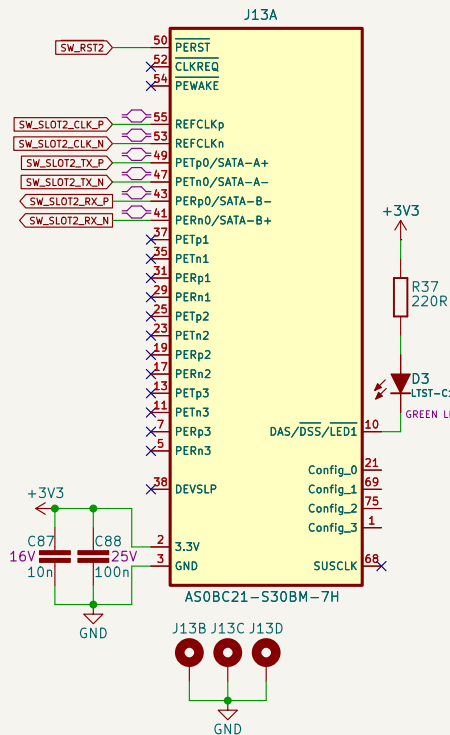
SW_EP0



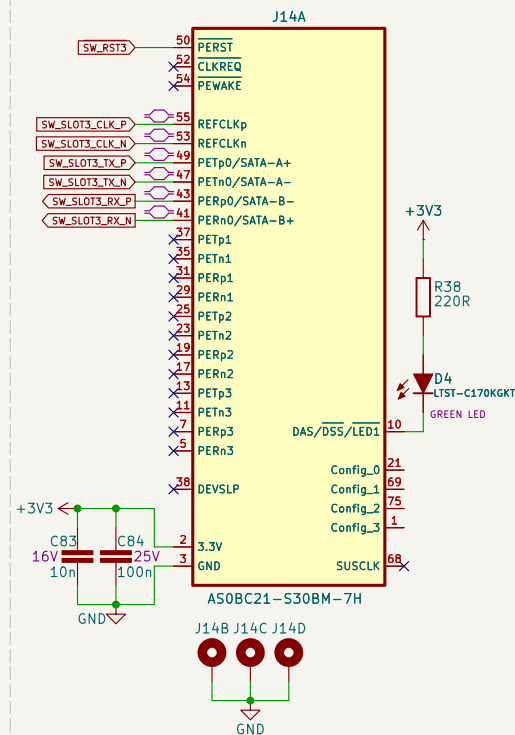
SW_EP1



SW_EP2



SW_EP3



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Sheet: /Switched 4 x 1-lanes PCIe/

File: 4_1-lanes_Pcle.kicad_sch

Title: openPCIE2 Backplane

Size: A4 Date: 2026-09-08

KiCad E.D.A. 10.0.6

Rev: r1B1

Id: 5/5